

1. The 1-D thermal diffusion equation is given by

$$\frac{\partial T(x, t)}{\partial t} = D \frac{\partial^2 T(x, t)}{\partial x^2}.$$

The thermal diffusion equation is a bit like a wave equation. Don't you think so?

Use the wave-like solutions of the form

$$T(x, t) \propto \exp[i(kx - \omega t)]$$

to solve this diffusion equation.

- (a) Show that

$$k = \pm(1 + i)\sqrt{\frac{\omega}{2D}}.$$

$$\frac{\partial T}{\partial t} = -i\omega T$$

$$\frac{\partial T}{\partial x} = ikT$$

$$\frac{\partial^2 T}{\partial x^2} = -k^2 T$$

$$-i\omega T = -Dk^2 T$$

$$k^2 = \frac{i\omega}{D}$$

$$k = \pm\sqrt{\frac{i\omega}{D}}$$

$$\sqrt{i} = \frac{1 + i}{\sqrt{2}}$$

$$k = \pm(1 + i)\sqrt{\frac{\omega}{2D}}$$

- (b) Find the general solution for the region $x \geq 0$.

$$T(x, t) \propto e^{i[(1+i)\sqrt{\frac{\omega}{2D}}x - \omega t]}$$

$$T(x, t) = e^{-i\omega t} \left(A e^{(1+i)\sqrt{\frac{\omega}{2D}}x} + B e^{-(1+i)\sqrt{\frac{\omega}{2D}}x} \right)$$

2. The radius and the thermal conductivity of a cylindrical wire is a and κ respectively. The wire's resistivity is ρ and it carries a uniform direct current I . The temperature of its surface is fixed at T_0 using a large water reservoir.

- (a) Show that the rate of heat generated per unit volume by the current carrying wire is

$$H = \rho j^2,$$

where j is the current density. Hint: Use the microscopic version of Ohm's law.

$$\begin{aligned} H &= j \cdot E \\ E &= \rho j \\ H &= j \cdot (\rho j) \\ &= \rho j^2. \end{aligned}$$

$$\boxed{H = \rho j^2}$$

(b) Show that the temperature $T(r)$ at radius r is given by

$$T(r) = \frac{\rho I^2}{4\pi^2 a^4 \kappa} (a^2 - r^2) + T_0.$$

$$\frac{\partial T}{\partial t} = D \nabla^2 T + \frac{H}{C}, \text{ where } D = \frac{\kappa}{C}$$

$$\frac{\partial T}{\partial t} = 0$$

$$\kappa \nabla^2 T = -H$$

$$\nabla^2 T = -\frac{H}{\kappa}$$

$$\nabla^2 T = \frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right)$$

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{H}{\kappa}$$

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{H}{\kappa} r$$

$$\int \frac{d}{dr} \left(r \frac{dT}{dr} \right) dr = \int -\frac{H}{\kappa} r dr$$

$$r \frac{dT}{dr} = -\frac{H}{2\kappa} r^2 + C_1$$

$$\frac{dT}{dr} = -\frac{H}{2\kappa} r + \frac{C_1}{r}$$

$$\int \frac{dT}{dr} dr = \int \left(-\frac{H}{2\kappa} r + \frac{C_1}{r} \right) dr$$

$$T(r) = -\frac{H}{4\kappa} r^2 + C_1 \ln r + C_2$$

$$C_1 = 0 \text{ because } \ln r \rightarrow -\infty \text{ as } r \rightarrow 0$$

$$T(r) = -\frac{H}{4\kappa} r^2 + C_2$$

$$T(a) = T_0$$

$$T_0 = -\frac{H}{4\kappa} a^2 + C_2$$

$$C_2 = T_0 + \frac{H}{4\kappa} a^2$$

$$T(r) = \frac{H}{4\kappa} (a^2 - r^2) + T_0$$

$$H = \rho j^2, j = \frac{I}{A} = \frac{I}{\pi a^2}$$

$$H = \rho \frac{I^2}{\pi^2 a^4}$$

$$T(r) = \frac{\rho I^2}{4\pi^2 a^4 \kappa} (a^2 - r^2) + T_0$$

- (c) If the wire is now placed in air at temperature T_{air} and the wire loses heat from its surface according to Newton's law of cooling, so that the heat flux from the surface of the wire is given by

$$\alpha[T(a) - T_{\text{air}}],$$

where α is a constant, find the temperature $T(r)$.

$$\begin{aligned} J &= -\kappa \frac{dT}{dr} \\ -\kappa \frac{dT}{dr} \Big|_{r=a} &= \alpha[T(a) - T_{\text{air}}] \\ \frac{dT}{dr} &= -\frac{H}{2\kappa} r \\ -\kappa T'(a) &= \alpha[T(a) - T_{\text{air}}] \\ -\kappa \left(-\frac{H}{2\kappa} a \right) &= \alpha \left[-\frac{H}{4\kappa} a^2 + C_2 - T_{\text{air}} \right] \\ \frac{H}{2} a &= \alpha \left[-\frac{H}{4\kappa} a^2 + C_2 - T_{\text{air}} \right] \\ \frac{Ha}{2\alpha} &= -\frac{H}{4\kappa} a^2 + C_2 - T_{\text{air}} \\ C_2 &= T_{\text{air}} + \frac{H}{4\kappa} a^2 + \frac{Ha}{2\alpha} \\ T(r) &= -\frac{H}{4\kappa} r^2 + T_{\text{air}} + \frac{H}{4\kappa} a^2 + \frac{Ha}{2\alpha} \\ T(r) &= \frac{H}{4\kappa} (a^2 - r^2) + T_{\text{air}} + \frac{Ha}{2\alpha} \end{aligned}$$

$$\boxed{T(r) = \frac{\rho I^2}{4\pi^2 a^4 \kappa} (a^2 - r^2) + \frac{\rho I^2}{2\pi^2 a^3 \alpha} + T_{\text{air}}}$$